

IBRAHIM HAMID

Software Engineer | AI & Full-Stack Development

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PROFESSIONAL SUMMARY

Computer Science graduate and Software Engineer with hands-on experience building AI-powered, full-stack web applications. Skilled in Python (FastAPI), React, Next.js, Node.js, LangChain, and retrieval-augmented generation (RAG), with practical exposure to LLM integration, vector search, and REST API development. Additional strength in software QA, test case design, and Agile bug tracking with Jira across four internships. Seeking a Software Engineer / AI Engineer role building production features in modern JavaScript and Python stacks.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, HTML5, CSS3

Frontend: React, Next.js, Tailwind CSS, Framer Motion, Responsive UI Development

Backend & APIs: FastAPI, Node.js, NestJS, REST API Design, API Integration, Postman

AI & Machine Learning: LangChain, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Ollama, Qwen2.5, GPT-4o-mini, Prompt Engineering, Speech-to-Text (Whisper), Text-to-Speech, Vector Search / Embeddings

Databases: MongoDB, MongoDB Atlas Vector Search

Blockchain: Smart Contracts, Hardhat, Elliptic Curve Cryptography (ECC)

QA & Testing: Manual Testing, Test Case Design, Functional Testing, Integration Testing, Usability Testing, Cross-Browser Testing, Bug Tracking

Tools & Practices: Git, GitHub, Jira, Postman, Agile / Scrum, CI/CD, Software Development Life Cycle (SDLC)

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Brandilya Group

July 2026 – Present

Islamabad, Pakistan | Creative & AI Product Agency

- Contributing to Createlya, an AI-powered presentation platform, building and testing features across a Next.js frontend and NestJS backend.
- Working with python-pptx to programmatically generate, edit, and validate PowerPoint (.pptx) decks from structured content and design templates.
- Supporting integration of LLM-driven slide generation and semantic retrieval over a large slide-design library using vector embeddings and MongoDB Atlas Vector Search.
- Writing test cases, reproducing defects, and verifying releases against requirements before client delivery.

Software Engineering Intern — NOVOTECH Solutions

July 2025 – August 2025

Islamabad, Pakistan

- Developed an intelligent chatbot agent using FastAPI, React (Next.js), Tailwind CSS, and LangChain, delivering multi-turn conversational support.
- Conducted functional and integration testing of core agent components, validating retrieval pipelines and external API handlers against user stories before deployment.
- Executed performance and usability testing on the chatbot UI/UX, analysing response latency and confirming cross-browser compatibility.

Software Engineering Intern — National Telecommunication Corporation

March 2024 – May 2024

Islamabad, Pakistan

- Built a Medical Slip Automation System using Java, HTML, and CSS, replacing a manual paper workflow and streamlining internal documentation.
- Gained hands-on experience in the Network Operations Center (NOC), monitoring and managing live network infrastructure.

Quality Assurance Intern — Mobilink Microfinance Bank Ltd

June 2023 – August 2023

Islamabad, Pakistan

- Performed manual testing and test case design on the Mobilink mobile banking app, verifying feature correctness and user experience across platforms.
- Tracked defects end-to-end in Jira from creation to closure, linking test results back to requirements to confirm full coverage and improve QA/development coordination.

PROJECTS

Createlya — AI Presentation Platform

Brandlya Group

Tech stack: Next.js, NestJS, Python (python-pptx), LLM APIs, MongoDB Atlas Vector Search, OnlyOffice

- Contributed to a three-panel presentation editor combining a client content panel, a searchable slide-design library, and a live editing canvas.
- Worked on automated slide generation, where an LLM converts client briefs into structured, on-brand PowerPoint decks.
- Assisted with semantic slide search using vector embeddings so designers can retrieve relevant layouts from a large template library.

Chatbot-Agent — AI Conversational Assistant

Tech stack: Python (FastAPI), React (Next.js), LangChain, Ollama LLM, Tailwind CSS, Framer Motion

- Designed and implemented an AI-driven chatbot agent capable of multi-turn conversations with contextual awareness.
- Integrated Ollama (Qwen2.5) with LangChain for intent recognition, retrieval-augmented generation (RAG), and external API execution.
- Implemented voice support (Speech-to-Text and Text-to-Speech), built a responsive React frontend with Tailwind CSS and Framer Motion, and connected external APIs (weather, knowledge search) to extend agent capability.

Auto Market — Online Vehicle Marketplace

Tech stack: React, Node.js, MongoDB, AI APIs, Whisper

- Engineered a full-stack vehicle marketplace with a dynamic React frontend and a MongoDB-backed Node.js API.
- Integrated a custom AI chatbot using the Whisper model for Speech-to-Text, assisting users with platform navigation and vehicle inquiries.
- Connected multiple AI models and third-party APIs to streamline backend processes and improve the overall user experience.

Decentralized Hospital Record System

Tech stack: Blockchain, Solidity Smart Contracts, Hardhat, Elliptic Curve Cryptography (ECC)

- Developed a blockchain-based architecture for securely storing and managing decentralized hospital records.
- Engineered a custom managed-wallet system using ECC cryptography to authenticate and authorise doctors securely.
- Applied smart contracts and the Hardhat development framework to enforce data integrity and decentralized access control.

EDUCATION

Bachelor of Science in Computer Science

2022 – 2026 (Graduated)

Capital University of Science and Technology (CUST), Islamabad, Pakistan

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Google IT Support Professional Certificate (Coursera) | AI For Everyone, DeepLearning.AI (Coursera) | Android App Components (Coursera) | iOS App Development (Coursera)

VOLUNTEER EXPERIENCE

Volunteer Teacher — Education Health and Development Foundation

February 2025

- Taught English, Mathematics, and General Knowledge to primary school students from underprivileged backgrounds and mentored them to build learning motivation.